



Arthrology

COURSE OUTLINE

- ▶ Introduction
- ▶ Classification of Joints
- ▶ Fibrous Joint
- ▶ Cartilaginous Joint
- ▶ Synovial Joint
- ▶ Injuries of Joints

Introduction

- ▶ Joints or articulations are sites where two or more bones meet
- ▶ Functions
 - ▶ provide skeletal mobility
 - ▶ hold the skeleton together
- ▶ **Weakest parts** of the skeleton but have ability to resist the forces that tear them apart

Classification of Joints

- ▶ Structural classification
 - ▶ focuses on the material binding the bones together and whether or not there is a joint cavity
- ▶ Functional classification
 - ▶ based on the amount of movement allowed at the joint

Functional Classification

- ▶ **Synarthroses**
 - ▶ Immovable joints
- ▶ **Amphiarthroses**
 - ▶ Slightly movable joints
- ▶ **Diarthroses**
 - ▶ Freely movable joints

Structural Classification

- ▶ Fibrous
 - ▶ Joined by fibrous tissue
- ▶ Cartilaginous
 - ▶ Joined by cartilage
- ▶ Synovial
 - ▶ Joined and surrounded by a joint cavity

Summary of Joint Classes

Fibrous joints

- ◆ Suture
- ◆ Syndesmoses
- ◆ Gomphosis

Cartilaginous joints

- ◆ Synchondroses
- ◆ Symphyses

Synovial

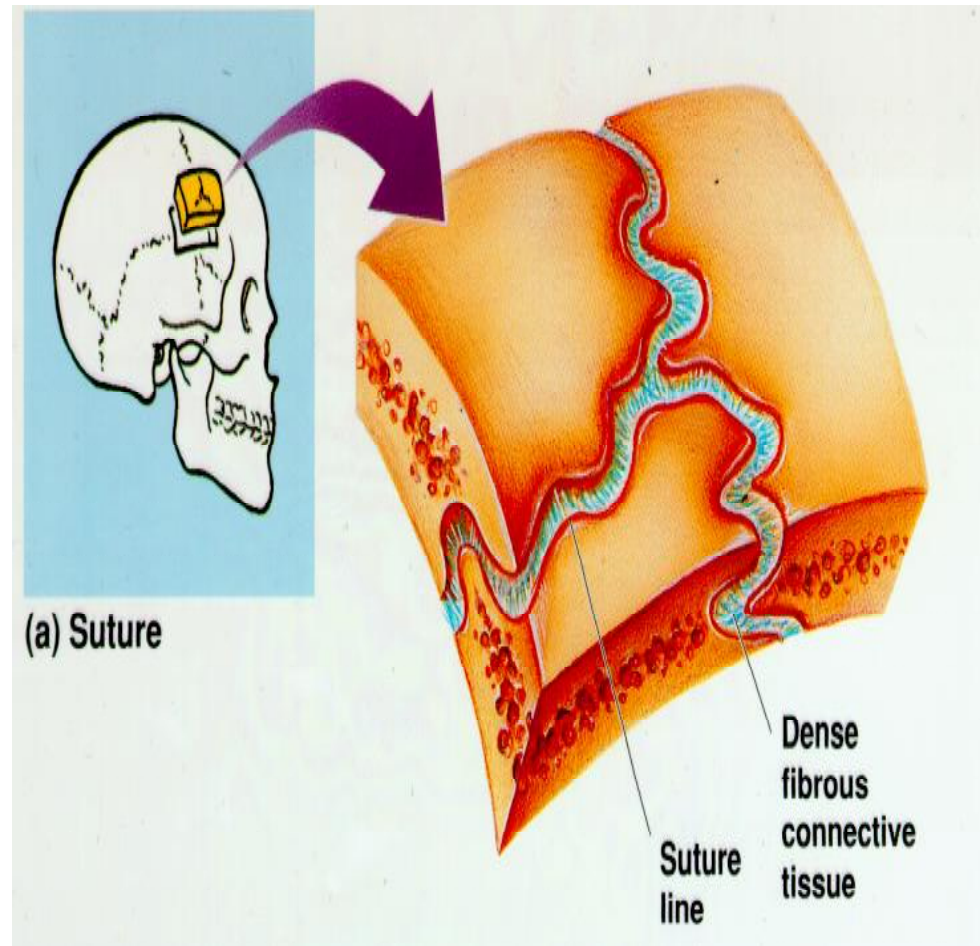
- ◆ Gliding
- ◆ Hinge
- ◆ Pivot
- ◆ Condylloid
- ◆ Saddle
- ◆ ball and socket

Fibrous Joints

- ▶ Bones are joined by fibrous tissue
- ▶ Types
 - ▶ Sutures
 - ▶ Dense fibrous connective tissue
 - ▶ Syndesmosis
 - ▶ A cord or band of connective tissue
 - ▶ Gomphosis
 - ▶ Peg-in-socket arrangement surrounded by fibrous tissue or periodontal ligament

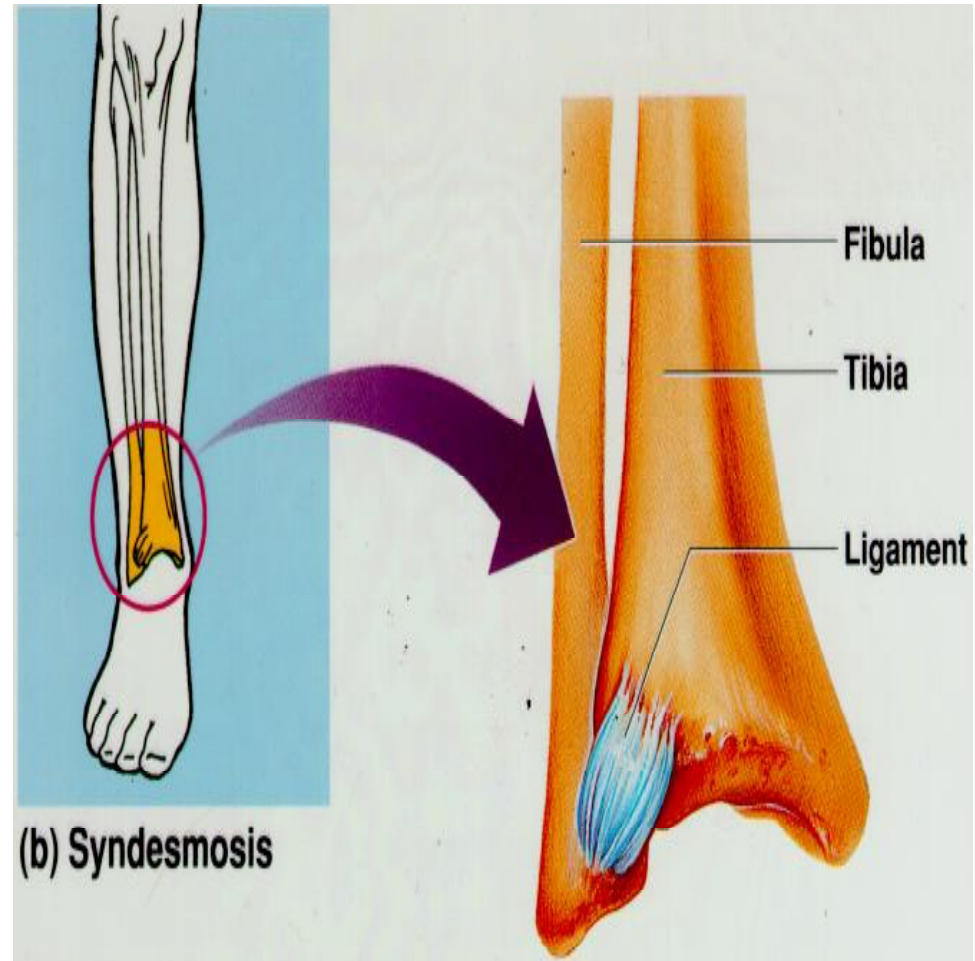
Suture

- ◆ Occurs only between bones of the skull
- ◆ Wavy articulating bone edges interlock
- ◆ Junction is filled by connective tissue



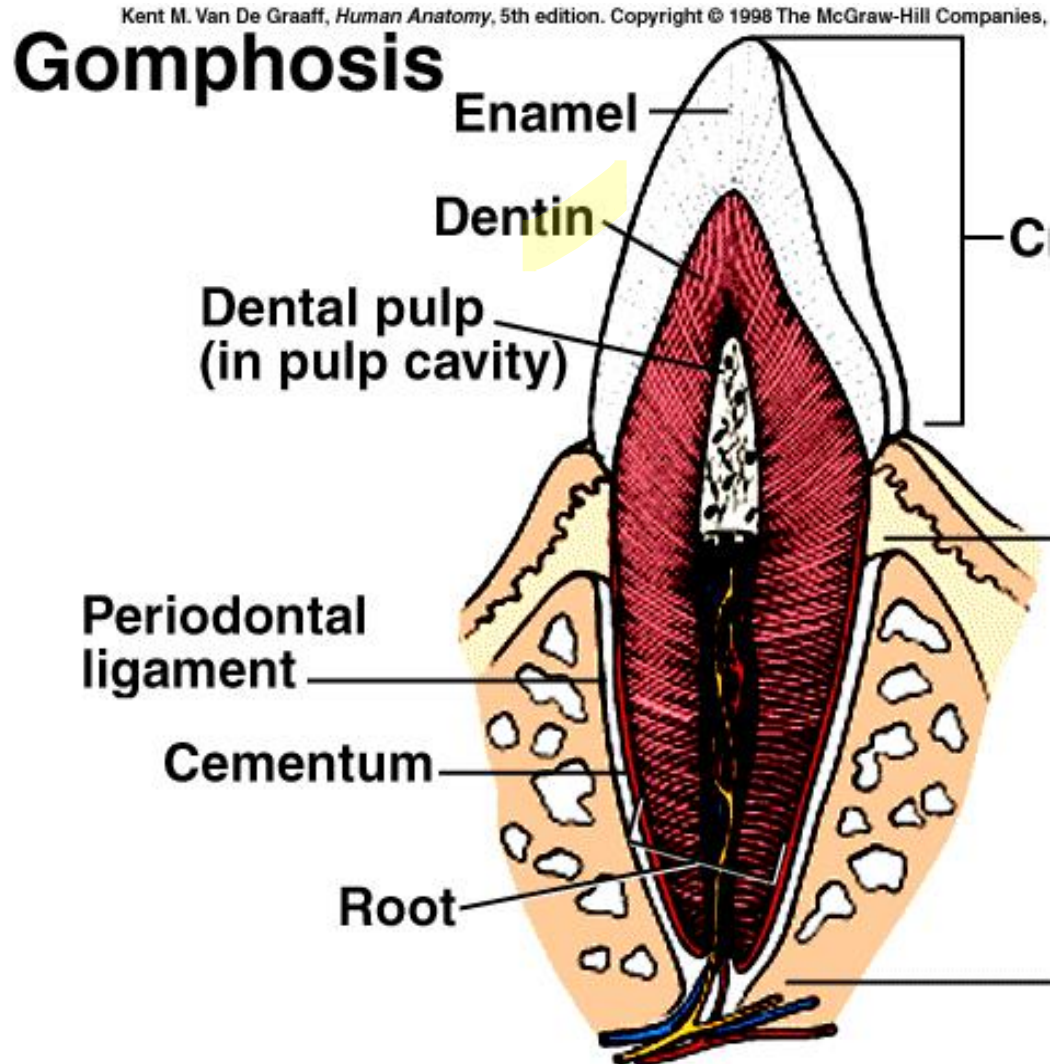
Syndesmosis

- ▶ bones are united by sheet of fibrous tissue
- ▶ permits the joint to flex
- ▶ True movement is not possible



Gomphosis

- ▶ unique joint b/n a tooth & its socket
- ▶ Fibrous tissue holds teeth in their sockets

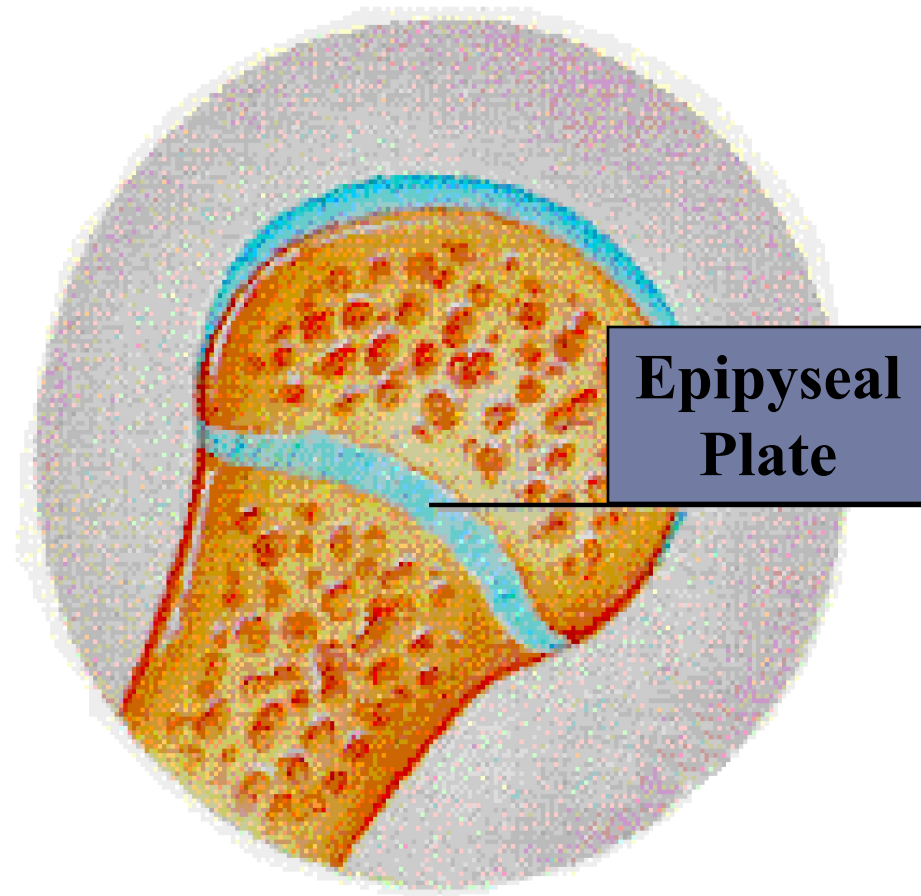


Cartilaginous Joints

- ▶ The articulating bones are united by cartilage
- ▶ Types
 - ▶ Synchondroses
 - ▶ Hyaline cartilage unites the bones
 - ▶ Symphyses
 - ▶ Fibrocartilage unites the bones

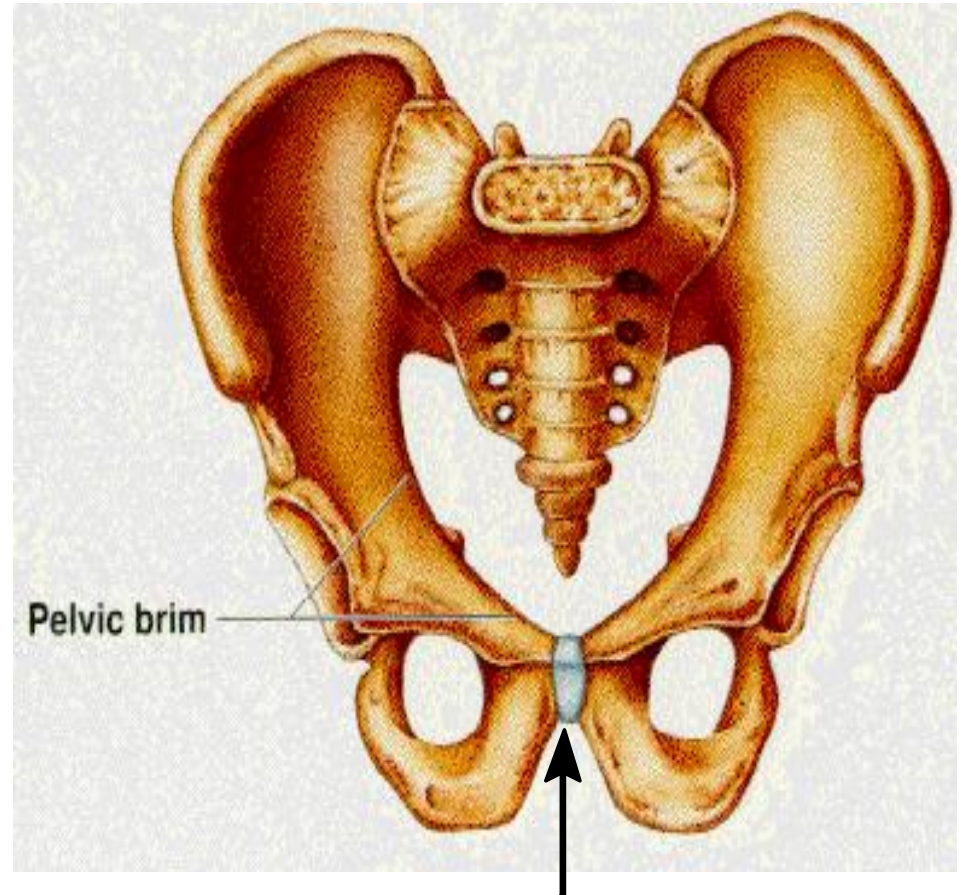
Synchondroses

- ▶ Hyaline cartilage unites the bones
- ▶ E.g.. **Epiphyseal plates** in growing children
- ▶ Provide for bone growth
- ▶ When growth ends all synchondroses become immovable



Symphyses

- ▶ Bone surfaces are covered with articular hyaline cartilage which is fused to a pad of **fibrocartilage**
- ▶ Fibrocartilage is resilient and acts as **a shock absorber** and permits limited movement



Pubic Symphys

Synovial Joints

- ▶ Articulating bones are located within a fluid containing joint cavity
- ▶ Permit substantial range of motion

Structures of Synovial Joint

- ▶ Articular cartilage
 - ▶ Hyaline cartilage on opposing bone surfaces
- ▶ Joint (synovial) cavity
 - ▶ Space filled with fluid
- ▶ Articular capsule
 - ▶ Capsule to confine fluid
- ▶ Synovial fluid
 - ▶ Fluid to lubricate joints
- ▶ Reinforcing ligaments
 - ▶ Maintain joint alignment

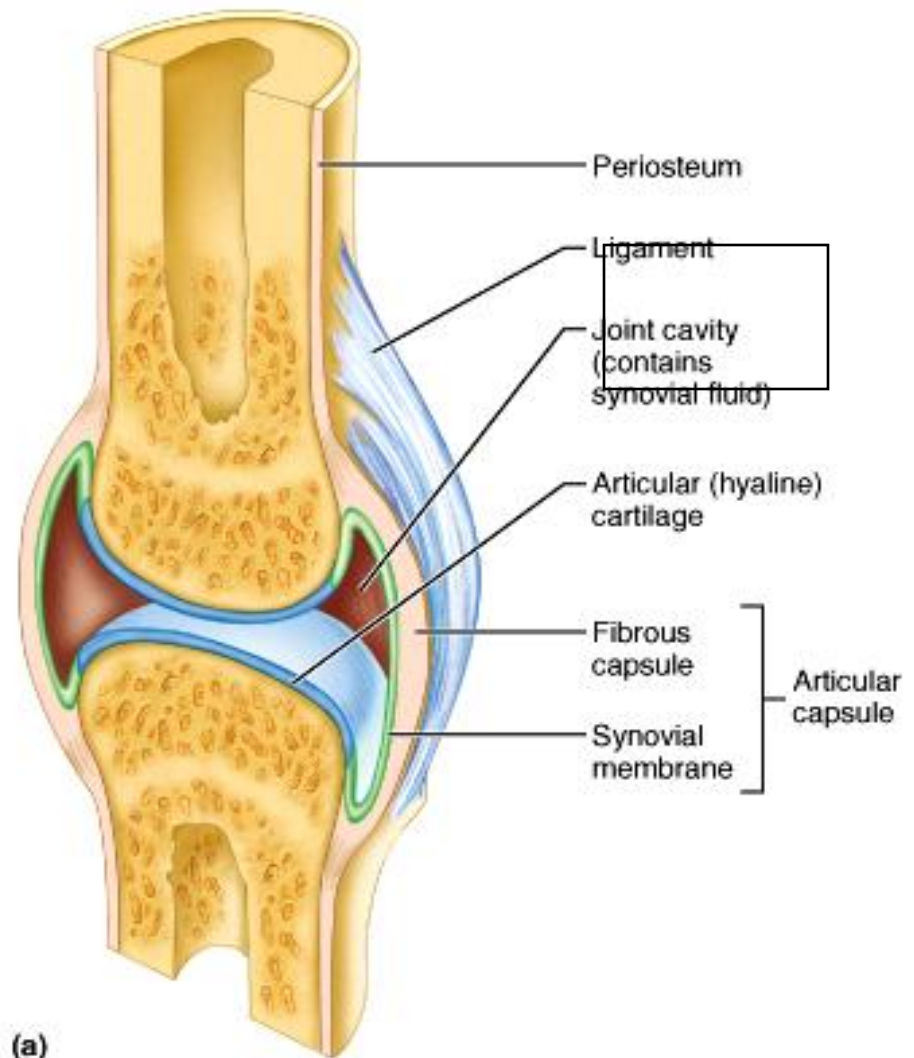
Articular Cartilage

- ▶ Hyaline cartilage covers the bone surfaces
- ▶ Cartilage absorbs the compression placed on the joint
- ▶ Cartilage keeps the bone ends from being crushed



Synovial cavity

- ▶ unique to synovial joints
- ▶ filled with synovial fluid

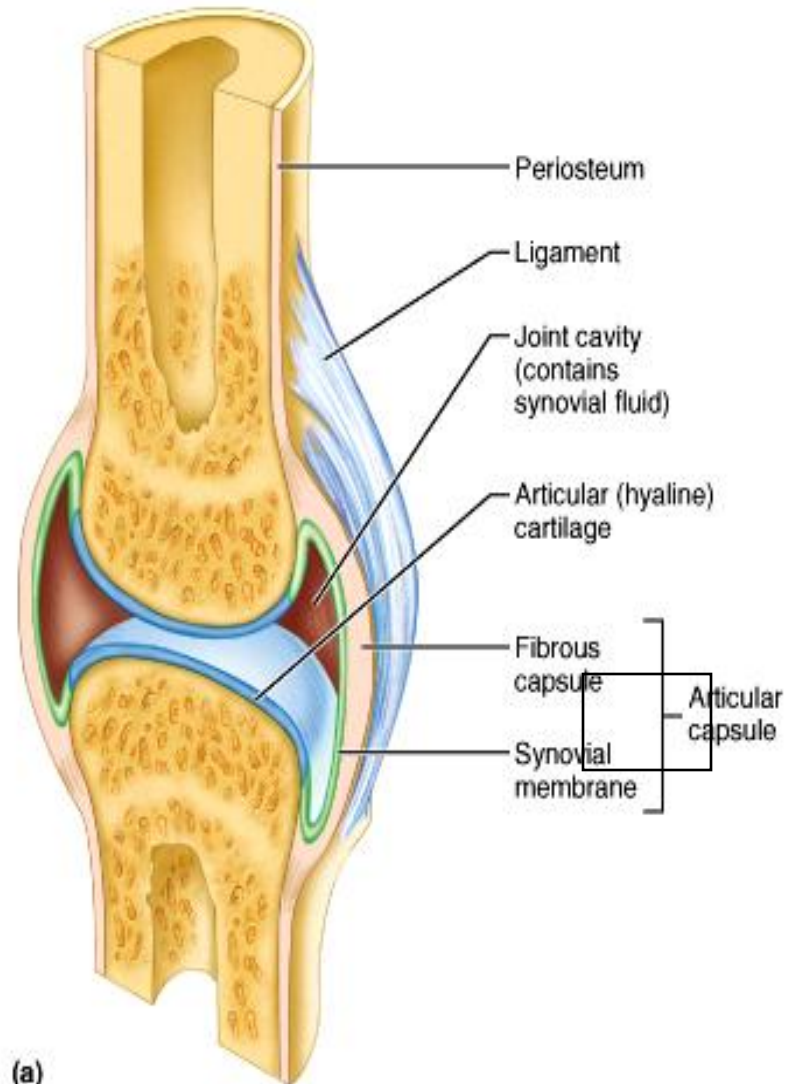


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Articular capsule

- ▶ The joint cavity is enclosed by a double layered articular capsule
- ▶ The external layer is a tough flexible fibrous capsule
- ▶ The inner synovial membrane



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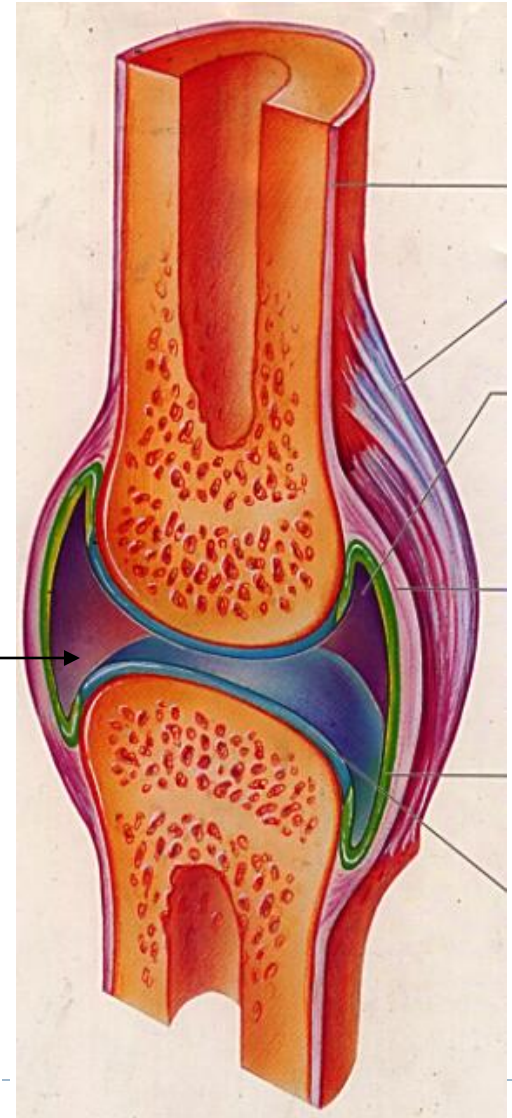
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Synovial Fluid

- ▶ lubricates joint
- ▶ nourishing cells

**Synovial
Fluid**

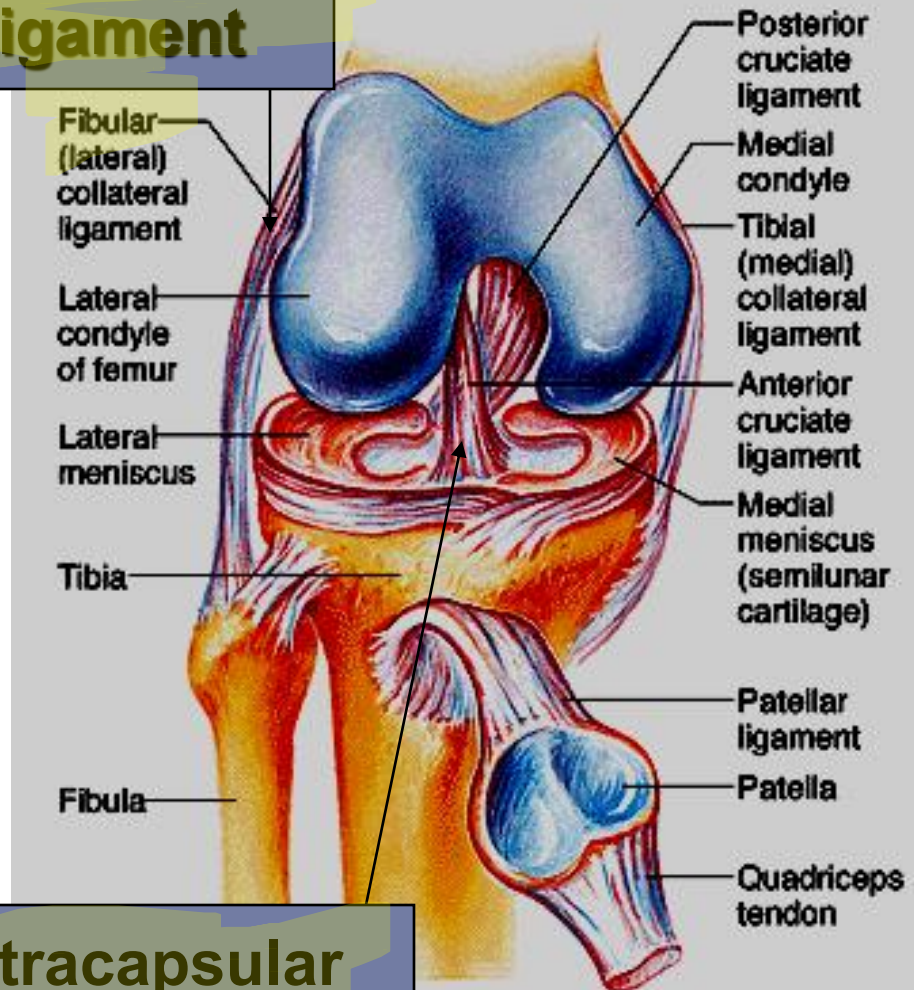


Reinforcing ligaments

- ▶ Ligaments reinforce joint

- ▶ Extra capsular
- ▶ Intra capsular

Extracapsular Ligament



Intracapsular Ligament

Synovial Joints

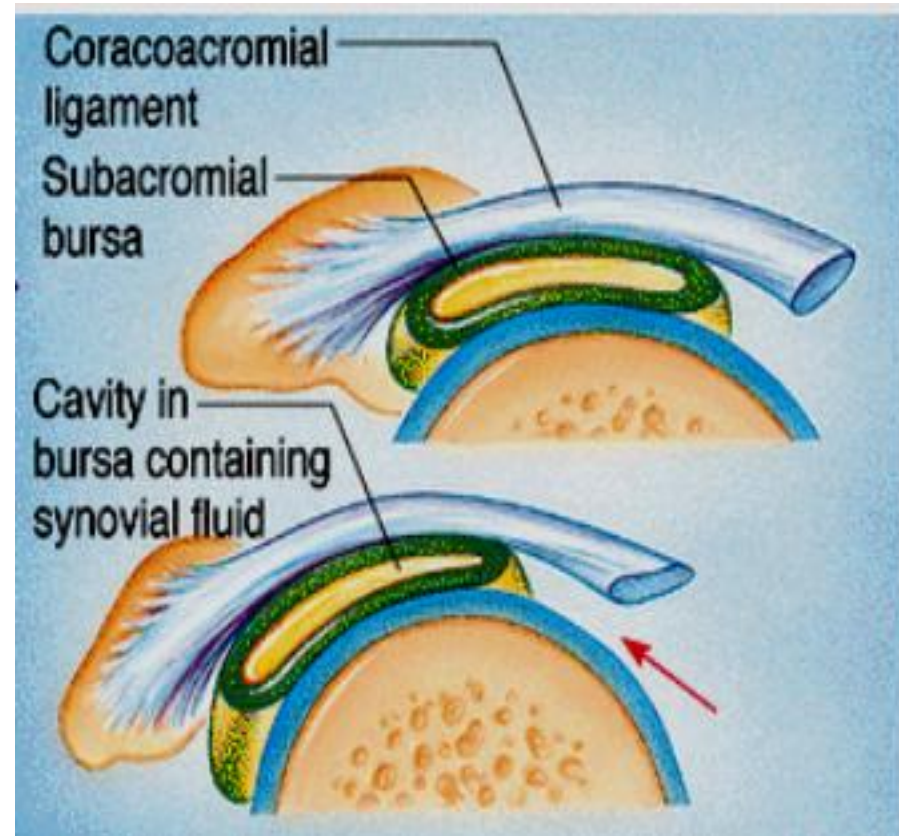
- ▶ Certain synovial joints **have additional structural features**
 - ▶ Fatty pads cushion the knee and hip joints
 - ▶ Fibrocartilage articular discs separates articular surfaces (menisci)
 - ▶ Articular discs improve the fit between the articulating surface

Bursae and Tendon Sheaths

- ▶ Bursae and tendon sheaths are closely associated with synovial joints
- ▶ Essentially sacs of lubricant
- ▶ Function: reduce friction between adjacent structures

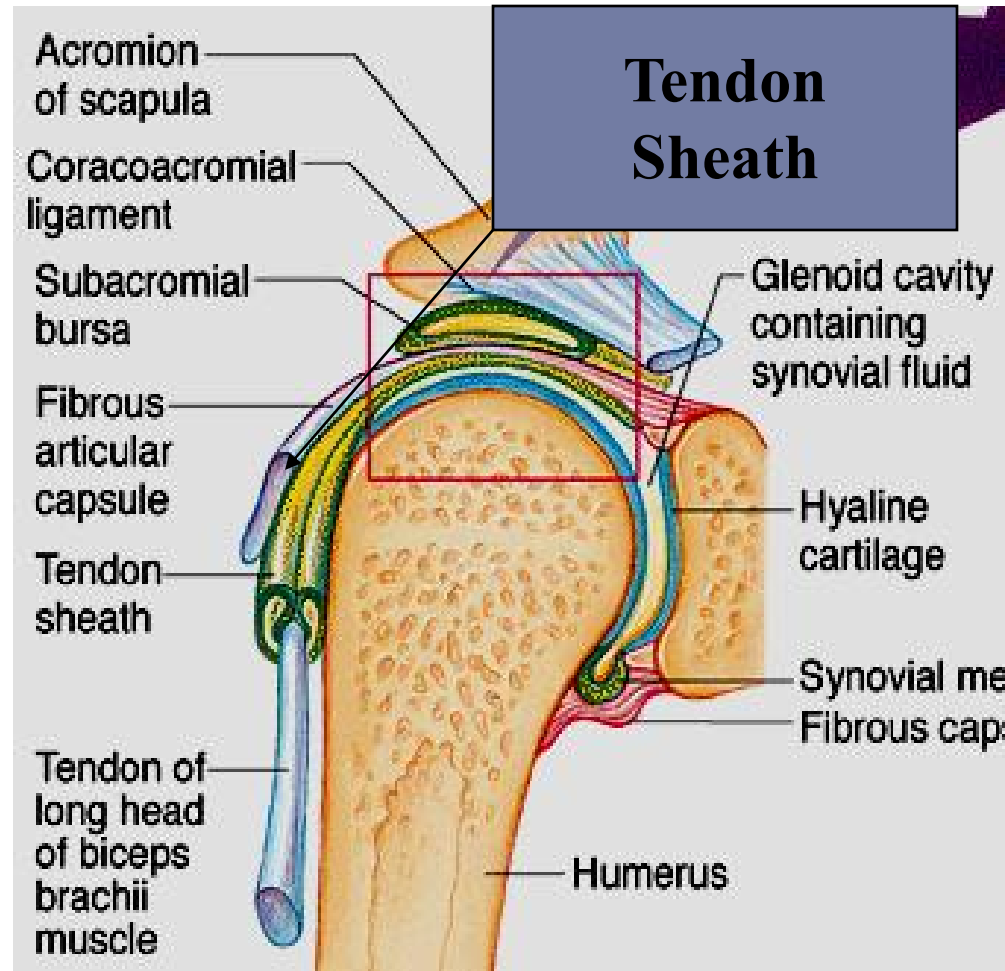
Bursae

- ▶ are flattened fibrous sacs lined with synovial membrane and containing a thin film of synovial fluid
- ▶ Common at sites where ligaments, skin, muscles or tendons rub against a bone

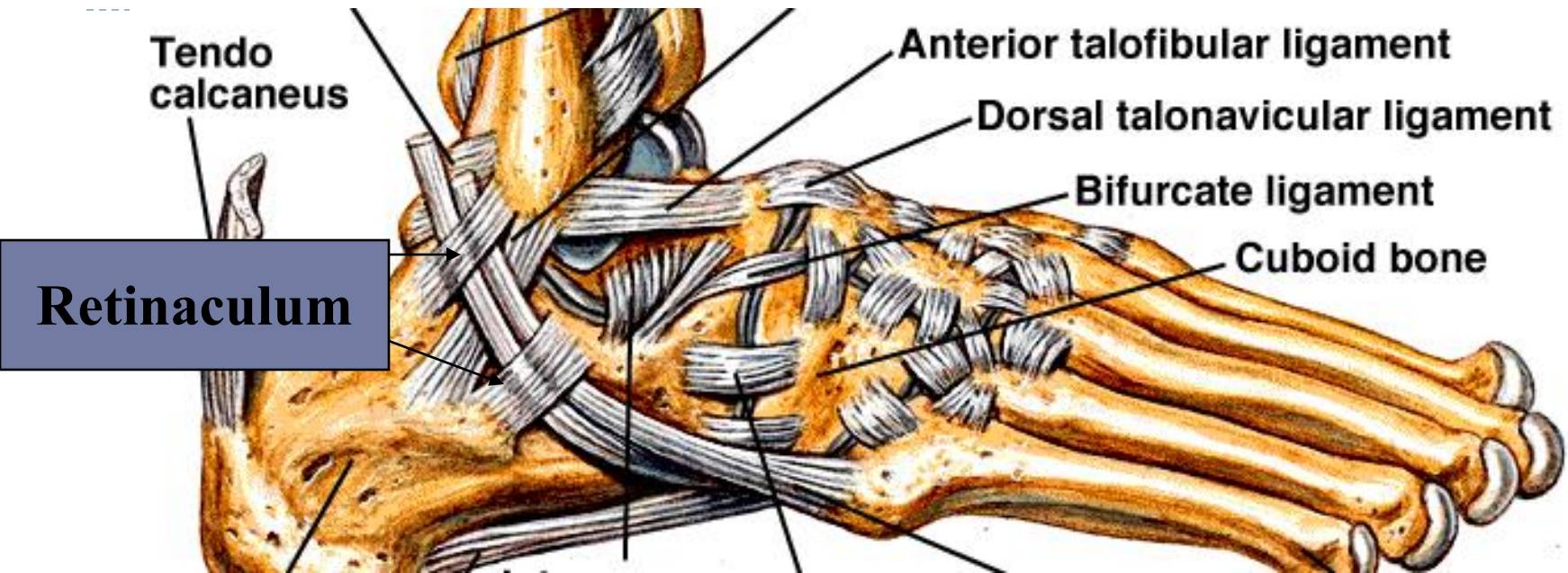


Tendon Sheaths

- An elongated bursa that wraps completely around a tendon subjected to friction.
- Tendon slides within this lubricated sleeve
- Common at sites where the tendon is subject to friction from other tendons or bone features



Retinaculum



- Retinaculum function to confine tendons to a specific line of pull
- Similar to a pulley or gear changing the angle of force exerted by a machine

Factors Influencing Synovial Joint Stability

- ▶ The nature of the articular surfaces
- ▶ The number and positioning of the ligaments
- ▶ The tone and strength of the muscles acting upon the joint

Movements Allowed by Synovial Joints

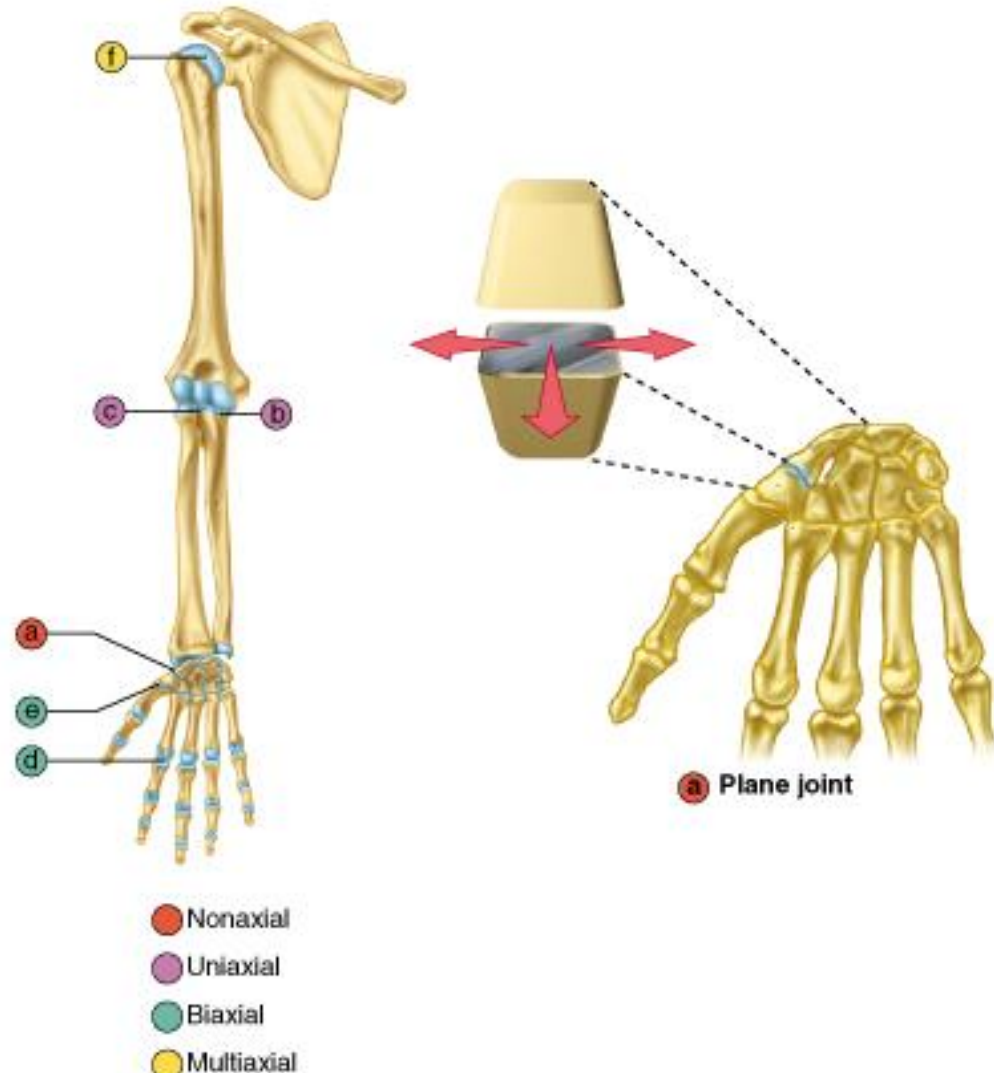
- ▶ Non-axial: no rotation around an axis
- ▶ Uniaxial: motion is within a single plane
- ▶ Biaxial: allow movement in two planes
- ▶ Multiaxial: movement is possible in all planes

Types of Synovial Joints

- ▶ Based on the shape of their articular surfaces there are six major categories of synovial joints
 - ▶ Plane (sliding)
 - ▶ Hinge
 - ▶ Pivot
 - ▶ Condylloid
 - ▶ Saddle
 - ▶ ball and socket

Plane Joint

- Articular surfaces are essentially flat
- Allow only short slipping or gliding movements
- nonaxial joint
- Examples
 - Intercarpals
 - Intertarsals
 - Vertebrae

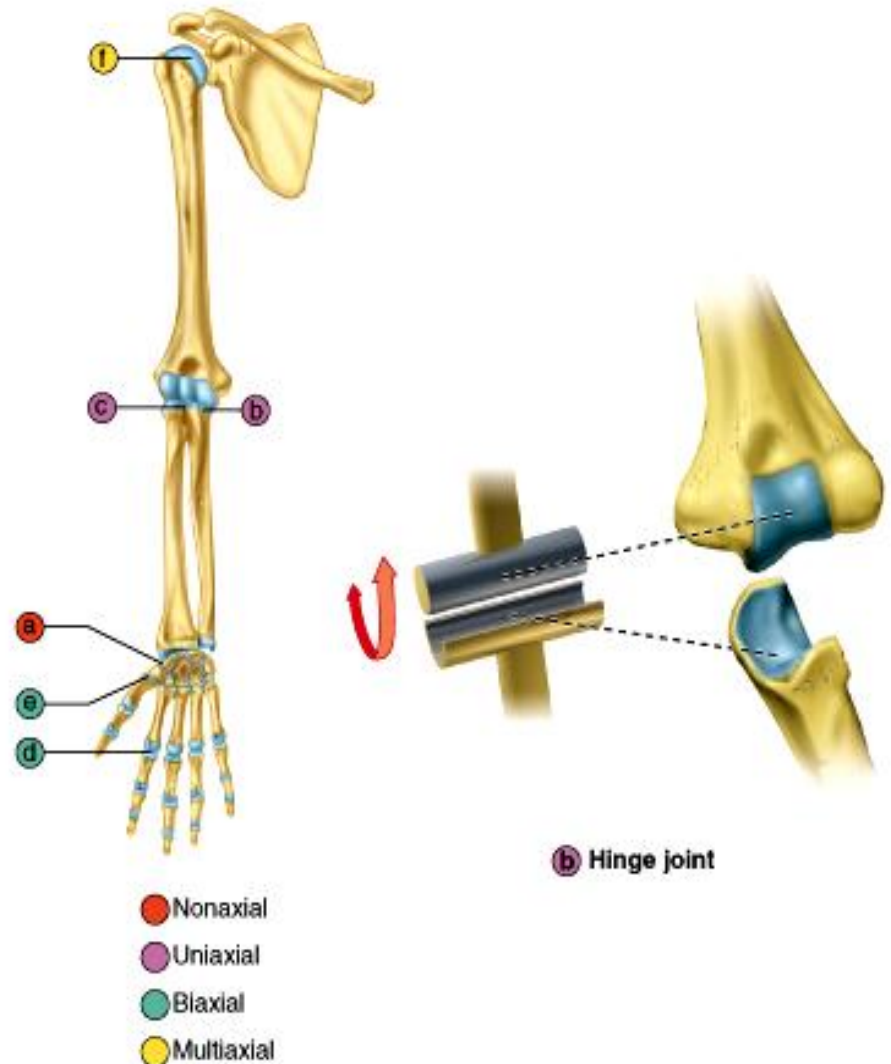


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Hinge Joint

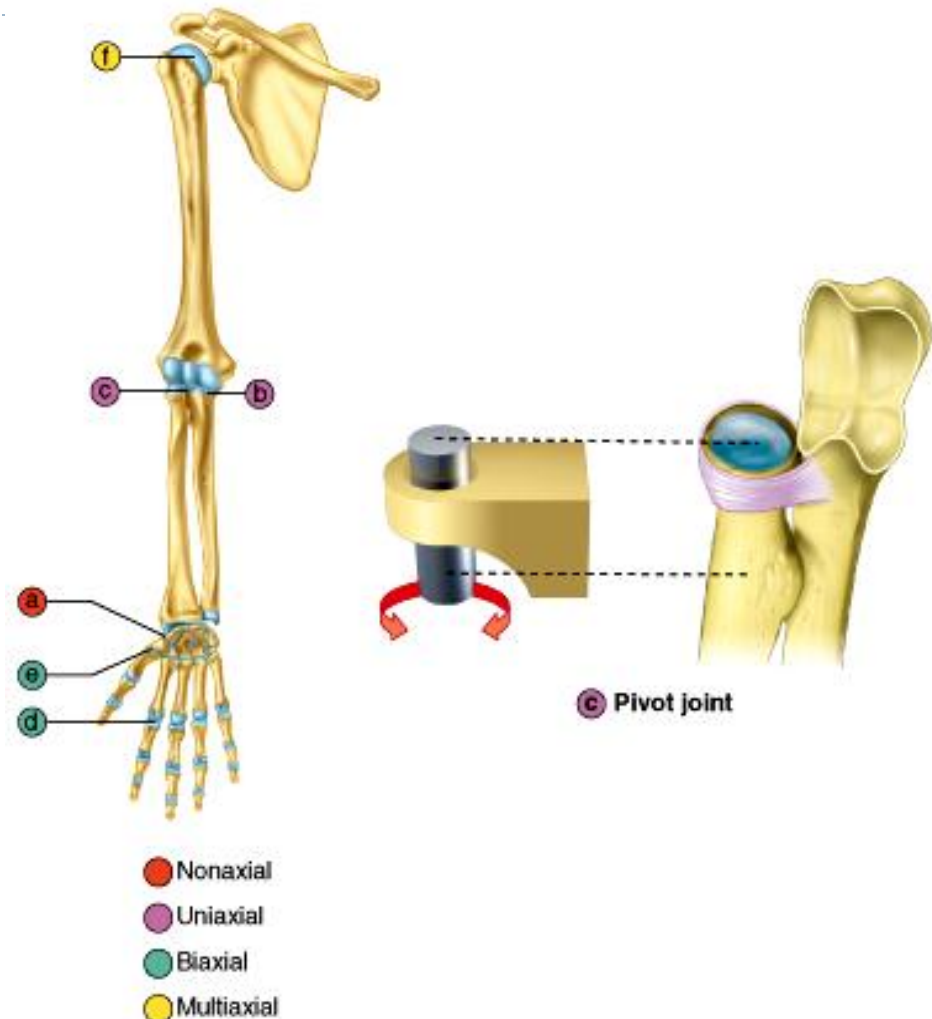
- ▶ a cylindrical shaped projection of bone fits into a trough shaped surface of another bone.
- ▶ Motion is within a **single plane (uniaxial)**
- ▶ Joint components resemble that of a mechanical hinge
- ▶ Example
 - ▶ **elbow joint**



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Pivot Joint

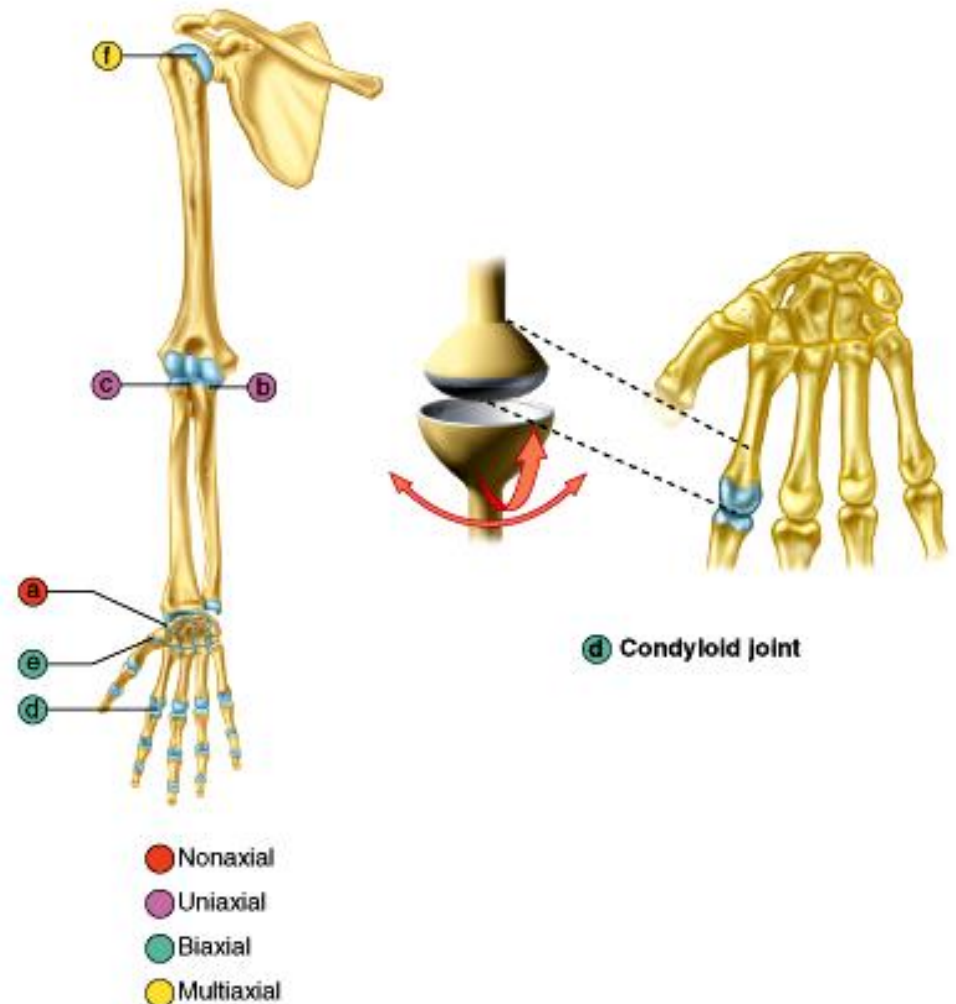
- ▶ The rounded end of a bone protrudes into a ring of bone and ligaments on another bone
- ▶ Only movement allowed is rotation of bone around long axis
- ▶ Example
 - ▶ the joint between the atlas and axis.
 - ▶ proximal radioulnar joint



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Condyloid Joints

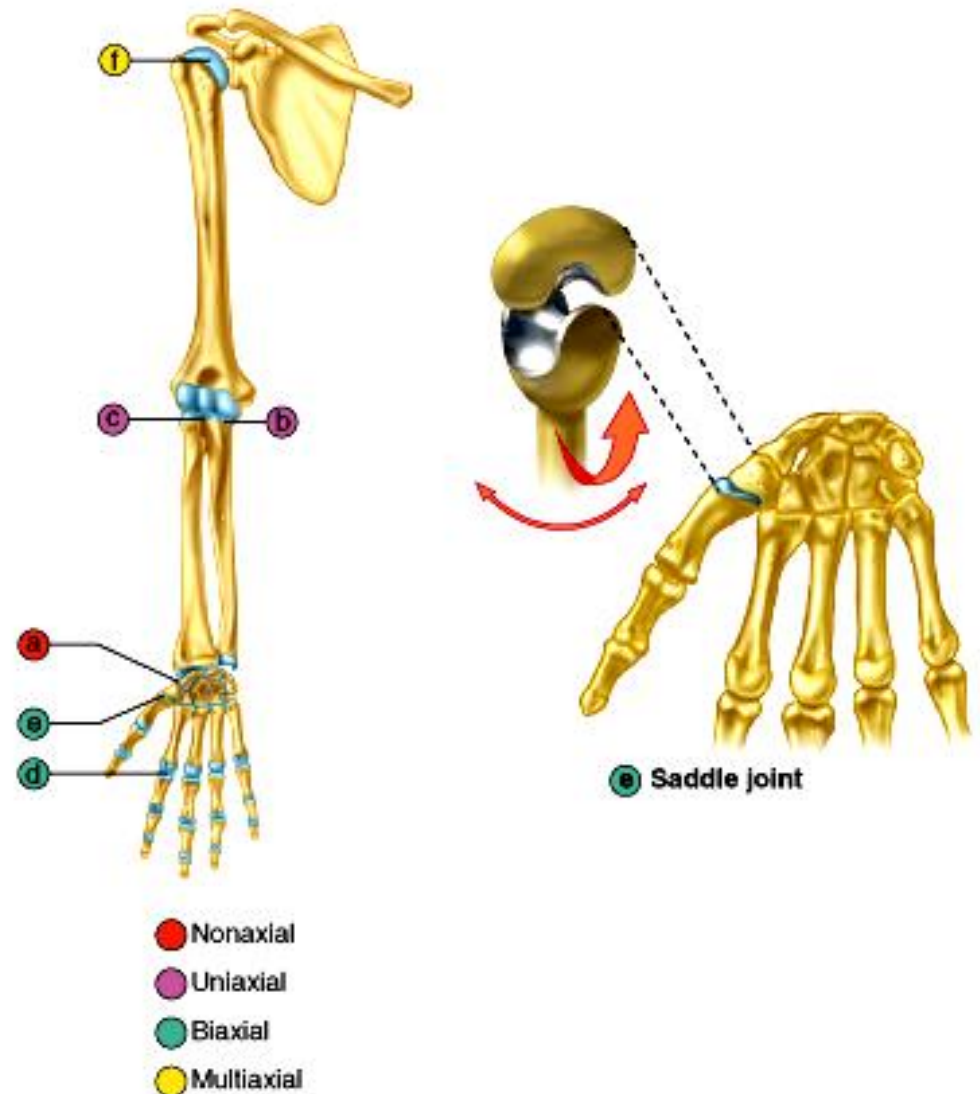
- ▶ The **oval articular** surface of one bone fits into a complementary **concavity** in another
- ▶ Both articulating surfaces are oval shaped
- ▶ **biaxial joints**
- ▶ Example
 - ▶ **Metacarpo-phalangeal joints**



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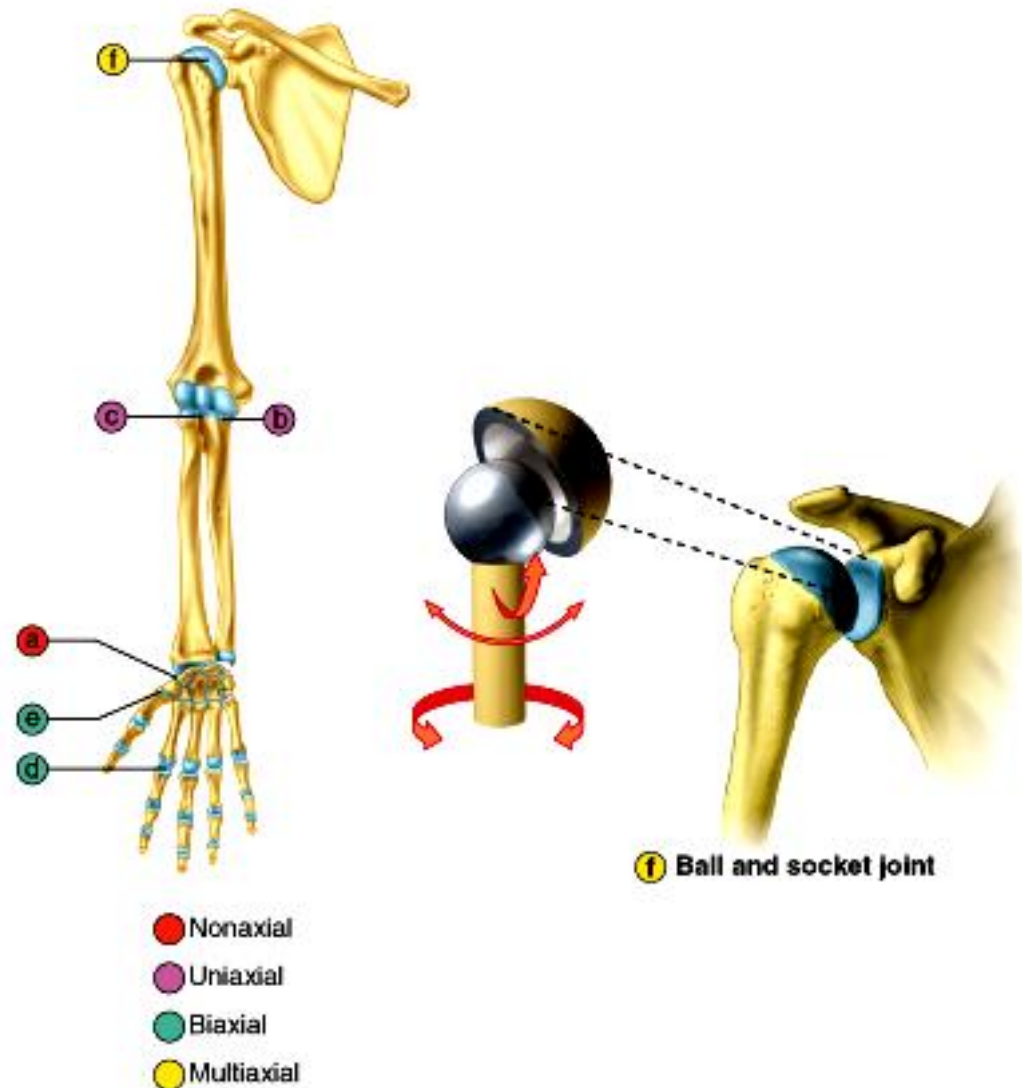
Saddle Joints

- ▶ Each surface has both a concave and a convex surface that fit together
- ▶ Example
 - ▶ carpometacarpal
- ▶ Biaxial movement



Ball and Socket Joint

- ▶ The **spherical head** of one bone articulates with the cuplike socket of another
- ▶ **Multiaxial**
- ▶ The most freely moving synovial joint
- ▶ Movements in all planes is allowed
- ▶ Examples
 - ▶ **Shoulder joint**
 - ▶ **Hip joint**



Injuries of Joints

- ▶ Sprains: **Ligament** supporting a joint are stretched or torn
- ▶ Strains: **Tendons** or **muscle fibers** are stretched or torn
- ▶ Cartilage tear: fragmentation of the cartilaginous tissue
- ▶ Dislocation: bones are forced out of their normal alignments at a joint
- ▶ Bursitis/Tendonitis: Inflammation caused by trauma or more frequently overuse